

CompTIA Security+ Exam SY0-701

Lesson 12



Explain Alerting and Monitoring
Concepts

Objectives

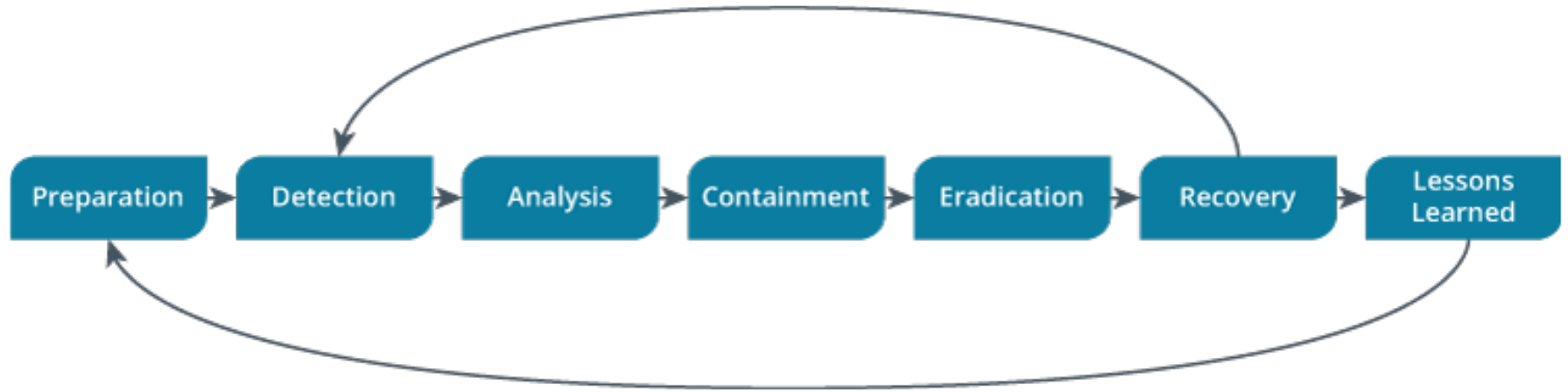
- Summarize incident response and digital forensics procedures
- Utilize appropriate data sources for incident investigations
- Explain security alerting and monitoring concepts and tools

Lesson 12

Topic 12A

Incident Response

Incident Response Processes



Preparation (1)

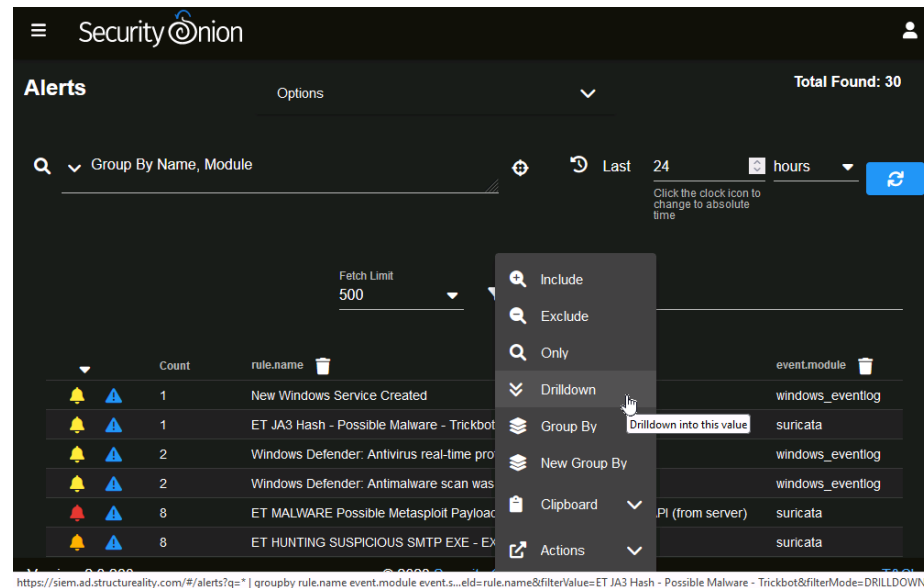
- Cybersecurity infrastructure
- Cyber Incident Response Team
 - Reporting, categorizing, and prioritizing (triage)
 - CIRT/CERT/CSIRT/SOC
 - Management/decision-making authority
 - Incident analysts
 - Roles beyond technical response
 - Legal, Human Resources (HR), Marketing

Preparation (2)

- Communication plan
 - Prevent inadvertent disclosure
 - Call list identifying trusted parties
 - Share data on a need to know basis
 - Out-of-band communications—avoid alerting intruder
- Stakeholder management
 - Communication with internal and external stakeholders
 - Notification and reporting
- Incident response plan (IRP)

Detection

- Detection channels
 - Monitoring and alerting from logs and other data sources
 - Deviation from baseline metrics
 - Manual inspection
 - Notification procedures
 - Public reporting and whistleblowing
- First responder
 - Member of CIRT taking charge of a reported incident



Analysis



- Analysis and incident identification
 - Classify and prioritize
 - Downgrade low priority alerts to log-only
- Impact
 - Data integrity, downtime
 - Economic/publicity
 - Scope
 - Detection time, recovery time
- Category
 - Kill chains and threat intelligence
- Playbooks

Containment

- Issues
 - Loss control
 - Countermeasures available
 - Preserving evidence
- Isolation-based containment
- Segmentation-based containment

Eradication and Recovery

- Reconstitution of affected systems
- Reaudit security controls
- Ensure that affected parties are notified

Lessons Learned

- Meet and report
- Root cause analysis
 - Five whys
 - Drill through incident with questions: Who, why, when, where, how, what
 - Walkthrough timeline

Testing and Training

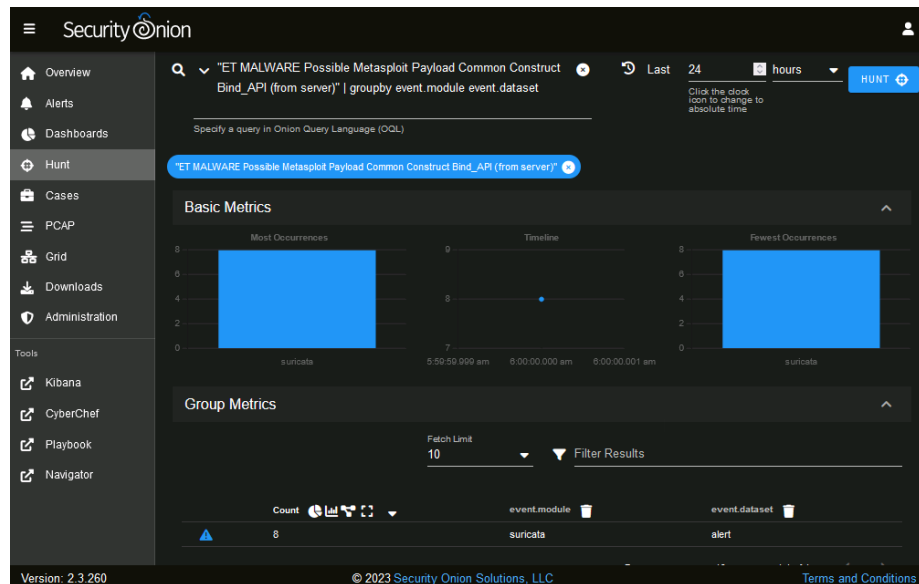


Members of the Kentucky and Alabama National Guard participate in a simulated network attack exercise.

(Photo by Kentucky National Guard Maj. Carla Raisler.)

- Testing
 - Tabletop
 - Facilitator presents a scenario
 - Does not involve live systems
 - Walkthroughs
 - Responders demonstrate response actions
 - Simulations
 - Red team performs a simulated intrusion
- Training

Threat Hunting



The Hunt dashboards in Security Onion can help to determine whether a given alert affects a single system only (as here), or whether it is more widespread across the network. (Screenshot courtesy of Security Onion securityonion.net.)

- Proactive process compared to reactive incident response
- Warning of new threat types
- Intelligence fusion and threat data
- Maneuver
 - Awareness that threat actor might take countermeasures

Review Activity: Incident Response

- Incident response processes
 - Preparation
 - Detection
 - Analysis
 - Containment
 - Eradication
 - Recovery
 - Lessons learned
- Testing and training
- Threat hunting

Lab Activity

- APPLIED LAB: Incident Response: Detection

Lesson 12

Topic 12B

Digital Forensics

Due Process and Legal Hold

- Digital forensics
 - Collecting evidence from computer systems to a standard that will be accepted in a court of law
- Evidence, documentation, and admissibility
 - Latent evidence
 - Collection must be documented
- Due process
 - Evidence collection and analysis procedures that ensure fairness
- Legal hold
 - Right to seize systems as evidence

Acquisition

- Legal seizure and search of devices
- Computer on/off state
- Order of volatility
 - CPU registers, cache memory, and non-persistent system memory (RAM)
 - Data on persistent storage
 - Remote logging and monitoring data
 - Physical configuration and network topology
 - Archival media

System Memory Acquisition

- Evidence recovery from non-persistent memory
 - Contents of temporary file systems, registry data, network connections, cryptographic keys, ...
- Live acquisition
 - Pre-install kernel driver

C:\Users\James\Downloads\Volatility 2.6 Win64 standalone.exe -t c:\dumps\memory.dmp --profile=Win7SP1x64_23418 pslist										
Volatility Foundation Volatility Framework 2.6										
Offset(V)	Name	PID	PPID	Thds	Hnds	Sess	Wow64	Start		Exit
0xffffffff8020a7040	System	4	0	106	632	-----	0	2020-01-09 21:20:03	UTC+0000	
0xffffffff8023d6d1d0	smss.exe	3088	4	2	29	-----	0	2020-01-09 21:20:03	UTC+0000	
0xffffffff80305f26a0	csrss.exe	396	388	8	370	0	0	2020-01-09 21:20:05	UTC+0000	
0xffffffff80304fa060	wininit.exe	432	388	3	75	0	0	2020-01-09 21:20:05	UTC+0000	
0xffffffff80306295e0	csrss.exe	444	424	8	293	1	0	2020-01-09 21:20:05	UTC+0000	
0xffffffff8030716b30	winlogon.exe	492	424	3	109	1	0	2020-01-09 21:20:05	UTC+0000	
0xffffffff80305fab30	services.exe	528	432	10	276	0	0	2020-01-09 21:20:05	UTC+0000	
0xffffffff8030732b30	lsass.exe	536	432	8	636	0	0	2020-01-09 21:20:05	UTC+0000	
0xffffffff803073db30	lsn.exe	544	432	10	142	0	0	2020-01-09 21:20:05	UTC+0000	
0xffffffff80307436a0	svchost.exe	652	528	10	349	0	0	2020-01-09 21:20:05	UTC+0000	
0xffffffff80307e66a0	svchost.exe	716	528	7	235	0	0	2020-01-09 21:20:05	UTC+0000	
0xffffffff80306566a0	svchost.exe	772	528	18	445	0	0	2020-01-09 21:20:05	UTC+0000	
0xffffffff80308bb060	svchost.exe	892	528	18	417	0	0	2020-01-09 21:20:06	UTC+0000	
0xffffffff80308fcb30	svchost.exe	936	528	32	940	0	0	2020-01-09 21:20:06	UTC+0000	
0xffffffff803093c060	svchost.exe	324	528	17	385	0	0	2020-01-09 21:20:06	UTC+0000	
0xffffffff8030960060	svchost.exe	744	528	15	379	0	0	2020-01-09 21:20:06	UTC+0000	
0xffffffff80309b7060	spoolsv.exe	1060	528	12	271	0	0	2020-01-09 21:20:06	UTC+0000	
0xffffffff80309dd060	svchost.exe	1096	528	19	316	0	0	2020-01-09 21:20:06	UTC+0000	
0xffffffff8030a58960	vmtoolsd.exe	1192	528	5	126	0	0	2020-01-09 21:20:06	UTC+0000	
0xffffffff8030a70b30	vmtoolsd.exe	1216	528	7	217	0	0	2020-01-09 21:20:06	UTC+0000	
0xffffffff8030a8c060	vmtoolsd.exe	1264	528	4	78	0	0	2020-01-09 21:20:06	UTC+0000	
0xffffffff8030accb30	vmtoolsd.exe	1296	528	5	92	0	0	2020-01-09 21:20:06	UTC+0000	
0xffffffff8030b32920	vmtoolsd.exe	1340	528	3	82	0	0	2020-01-09 21:20:07	UTC+0000	
0xffffffff80302ab3210	svchost.exe	1436	528	10	179	0	0	2020-01-09 21:20:07	UTC+0000	
0xffffffff8030bcf800	svchost.exe	1528	528	3	43	0	0	2020-01-09 21:20:08	UTC+0000	
0xffffffff8030c963a0	svchost.exe	1816	528	5	99	0	0	2020-01-09 21:20:08	UTC+0000	
0xffffffff8030ac5b30	svchost.exe	1976	528	14	323	0	0	2020-01-09 21:20:10	UTC+0000	
0xffffffff8030155b30	taskhost.exe	1964	528	9	157	1	0	2020-01-09 21:20:14	UTC+0000	
0xffffffff80301c3830	sppsvc.exe	2072	528	7	158	0	0	2020-01-09 21:20:14	UTC+0000	
0xffffffff8030260600	dmw.exe	2352	892	3	70	1	0	2020-01-09 21:20:18	UTC+0000	
0xffffffff8030238060	explorer.exe	2376	2344	24	784	1	0	2020-01-09 21:20:18	UTC+0000	
0xffffffff803033a2b30	lsuched.exe	2520	2456	8	233	1	1	2020-01-09 21:20:18	UTC+0000	
0xffffffff8030ba0b30	SearchIndexer.exe	2568	528	11	656	0	0	2020-01-09 21:20:24	UTC+0000	
0xffffffff803026a060	procp64.exe	2900	2376	8	382	1	0	2020-01-09 21:20:45	UTC+0000	
0xffffffff80306406a0	WmiPrvSE.exe	3024	652	7	118	0	0	2020-01-09 21:20:51	UTC+0000	
0xffffffff8030703190	Tcpv6.exe	916	2376	6	139	1	1	2020-01-09 21:21:27	UTC+0000	
0xffffffff802839b30	salter.exe	1808	2376	6	134	1	1	2020-01-09 21:23:49	UTC+0000	
0xffffffff803818230	WMIADAP.exe	380	936	5	85	0	0	2020-01-09 21:24:08	UTC+0000	

Screenshot: Volatility Framework volatilityfoundation.org.)

Disk Image Acquisition

```
root@kali:~# dcfldd if=/dev/sda hash=sha256 of=/root/FORENSIC/ROGUE.dd bs=512 co
nv=noerror
134217728 blocks (65536Mb) written.Total (sha256): 7a72be231f393d40e0ac72c62b3a7
3798f29f0ca7e0e279b8aececa291a34137

134217728+0 records in
134217728+0 records out
root@kali:~# sha256sum /dev/sda
7a72be231f393d40e0ac72c62b3a73798f29f0ca7e0e279b8aececa291a34137 /dev/sda
root@kali:~#
```

Using dcfldd (a version of dd with additional forensics functionality created by the DoD) and generating a hash of the source-disk data (sda).

- Non-volatile storage media and devices
- Acquisition types
 - Live acquisition
 - Static acquisition by shutting down the host
 - Static acquisition by pulling the plug
- Imaging utilities
 - Forensic software suites and file formats

Preservation

- Timeline and provenance
 - Record process of evidence acquisition
 - Use a write blocker
- Evidence integrity and non-repudiation
 - Cryptographic hashing and checksums
 - Take hashes of source device, reference image, and copy of image for analysis
- Chain of custody
 - Integrity and proper handling of evidence from collection, to analysis, to storage, and finally to presentation
 - Secure tamper-evident bagging
 - Secure storage facility and protection against environmental hazards

Reporting

- Summarizes contents of the digital data
- Conclusions from the investigator's analysis
- Professional ethics
 - Analysis must be performed without bias
 - Analysis methods must be repeatable
 - Evidence must not be changed or manipulated
- E-discovery
 - Electronically Stored Information (ESI)
 - Identify and de-duplicate files and metadata and facilitate search and tagging
 - Protect access and make tamper-evident
 - Facilitate disclosure

Review Activity: Digital Forensics

- Due process and legal hold
- Acquisition
 - Order of volatility
- System memory acquisition
- Disk image acquisition
- Preservation
 - Evidence integrity, reference hash, chain of custody
- Reporting
 - E-discovery

Lab Activity

- APPLIED LAB: Performing Digital Forensics

Lesson 12

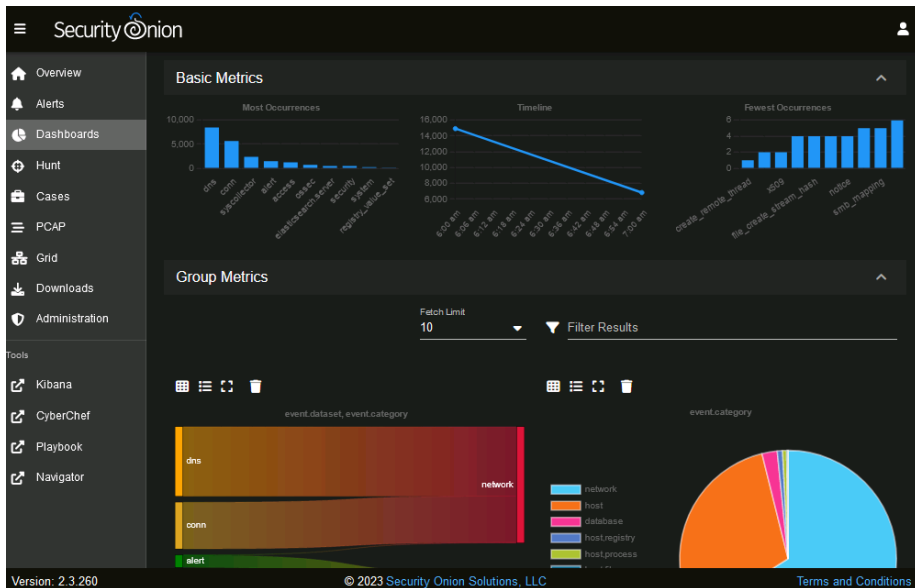
Topic 12C

Data Sources

Data Sources, Dashboards, and Reports (1)

- System memory and media device file system data and metadata
- Log files generated by network appliances
- Network traffic captured by sensors and/or intrusion detection systems
- Log files and alerts generated by network-based vulnerability scanners
- Log files generated by the OS components of client and server hosts
- Log files generated by applications and services running on hosts
- Log files and alerts generated by endpoint security software installed on hosts

Data Sources, Dashboards, and Reports (2)



Screenshot courtesy of Security Onion (securityonion.net.)

- Analyst dashboard
 - Console of alerts that require prioritization and investigation
- Manager dashboard
 - Overall status indicators
- Automated reports
 - Alerts and alarms
 - Status reports for response team, business managers, business owners, and compliance

Log Data

- Event
 - Format and source (local or redirected over network)
 - Event data versus metadata
- Windows Event Viewer
- Syslog
 - PRI – facility and severity
 - Header with timestamp and host
 - Message part

```
<5>Mar 12 05:11:40 LX1 kernel: [ 8399.702841] netfilter - ACCEPT
IN=eth0 OUT= MAC=00:15:5d:01:ca:55:00:15:5d:01:ca:ad:08:00 SRC=10.1.0.102 DST=10.1.0.10
LEN=88 TOS=0x00 PREC=0x00 TTL=128 ID=11507 DF PROTO=TCP SPT=1901 DPT=22 WINDOW=32767 RES=0x00 ACK PSH URGP=0
<5>Mar 12 05:11:46 LX1 kernel: [ 8404.945586] netfilter - ACCEPT
IN=eth0 OUT= MAC=00:15:5d:01:ca:55:00:15:5d:01:ca:ad:08:00 SRC=10.1.0.102 DST=10.1.0.10
LEN=52 TOS=0x00 PREC=0x00 TTL=128 ID=11510 DF PROTO=TCP SPT=1906 DPT=80 WINDOW=65535 RES=0x00 SYN URGP=0
<4>Mar 12 05:12:07 LX1 kernel: [ 8426.739265] netfilter - DROP
IN=eth0 OUT= MAC=00:15:5d:01:ca:55:00:15:5d:01:ca:ad:08:00 SRC=10.1.0.102 DST=10.1.0.10
LEN=52 TOS=0x00 PREC=0x00 TTL=128 ID=11613 DF PROTO=TCP SPT=1911 DPT=21 WINDOW=64240 RES=0x00 SYN URGP=0
```

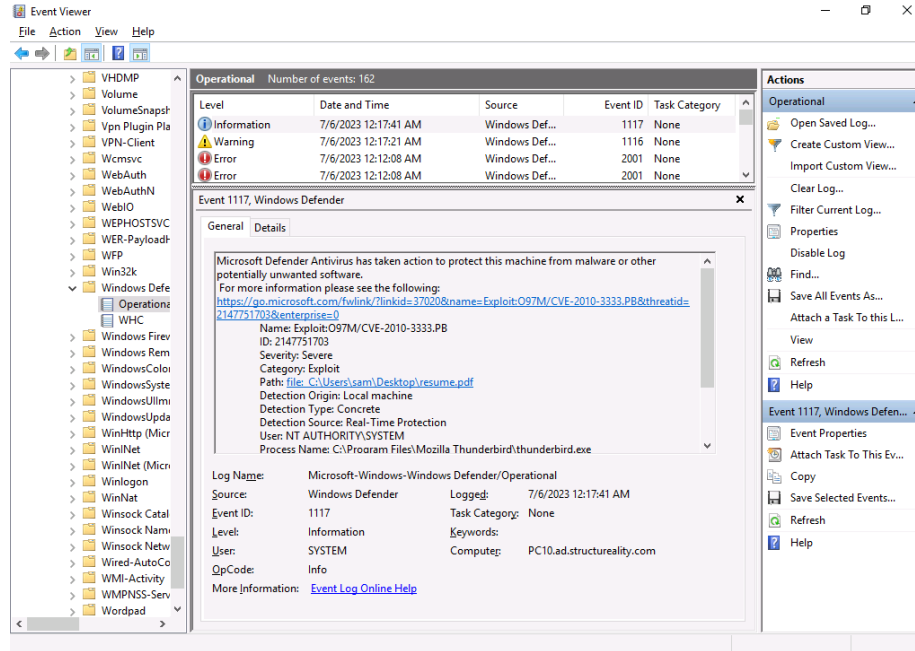
Host Operating System Logs

```
00:01:57 lamp sshd[453]: Server listening on 0.0.0.0 port 22.
00:01:57 lamp sshd[453]: Server listening on :: port 22.
00:17:01 lamp CRON[744]: pam_unix(cron:session): session opened for user root(uid=0) by (ui
00:17:01 lamp CRON[744]: pam_unix(cron:session): session closed for user root
00:26:47 lamp login[415]: pam_unix(login:auth): authentication failure; logname=LOGIN uid=0
00:26:51 lamp login[415]: FAILED LOGIN (1) on '/dev/tty1' FOR 'root', Authentication failur
00:26:59 lamp login[415]: FAILED LOGIN (2) on '/dev/tty1' FOR 'root', Authentication failur
00:27:05 lamp login[415]: FAILED LOGIN (3) on '/dev/tty1' FOR 'root', Authentication failur
00:27:11 lamp login[415]: pam_unix(login:session): session opened for user lamp(uid=1000) b
00:27:11 lamp systemd-logind[396]: New session 3 of user lamp.
00:27:11 lamp systemd: pam_unix(systemd-user:session): session opened for user lamp(uid=100
00:29:26 lamp sudo:      lamp : TTY=ttty1 ; PWD=/home/lamp ; USER=root ; COMMAND=/usr/bin/cat
00:29:26 lamp sudo: pam_unix(sudo:session): session opened for user root(uid=0) by lamp(uid
```

Linux authentication log showing SSH remote access is enabled, failed authentication attempts for root user, and successful login for lamp user.

- Security/audit logs
 - Authentication and authorization events
 - File system events
- Windows
 - Application, security, and system
- Linux
 - Syslog versus Journald
 - Common log files
- macOS unified logging

Application and Endpoint Logs



Windows Defender logging detection and quarantine of malware to Event Viewer.
(Screenshot used with permission from Microsoft.)

- Application logs
- Endpoint logs
 - Security software running on hosts
 - Summarize volume of detection incidents to indicate threat levels
 - Incident attribution and threat intelligence
- Vulnerability scans

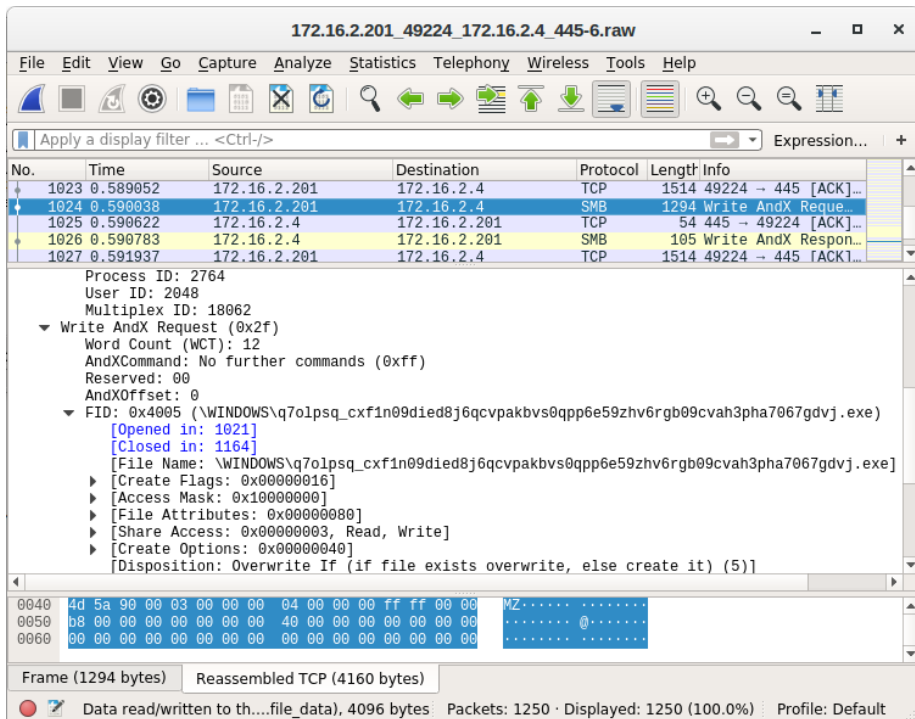
Network Data Sources

- Network logs
 - Traffic and access data from network appliances
- Firewall logs
 - Correlate to malicious activity on hosts
- IPS/IDS logs
 - Summarize volume of detection incidents to indicate threat levels
 - Attribution and threat intelligence

>		2023-07-24 07:18:56.181 -05:00	10.1.24.66	58300	10.1.16.2	25	ET HUNTING SUSP
▼		2023-07-24 06:49:34.036 -05:00	10.1.24.66	8443	10.1.16.2	1677	ET MALWARE Possi
	@timestamp	2023-07-24T11:49:34.036Z					
	destination.ip	10.1.16.2					
	destination.port	1677					
	ecs.version	8.0.0					
	event.category	network					
	event.dataset	alert					
	event.ingested	2023-07-24T11:49:38.931Z					
	event.module	suricata					
	event.severity	3					
	event.severity_label	high					
	message	{"timestamp":"2023-07-24T11:49:34.036379+0000","flow_id":"132699009128427","in_iface":"bond0","event_h ature_id":"2025644","rev":"1","signature":"","ET MALWARE Possible Metasploit Payload Common Construct Bind atacenter","internal","Internet","Perimeter"},"former_category":["TROJAN"],"signature_severity":["Critical"],"tag r"},"flow_from_server.established; content.V[60 89 e5 31]"," content.V[64 8b]"," distance:1; within:2; content:I ance:1; within:5; classtype:trojan-activity; sid:2025644; rev:1; metadata affected_product Any, attack_target aploit, updated_at 2018_07_10;"},"app_proto":"failed","payload_printable":""," December... AM. PM. MM/dd n..Feb..Mar..Apr..May..Jun..Jul..Aug..Sep..Oct..Nov..Dec..January..February...I s...e.n - U.S.....In.V..... I!#\$%&'()*+,-./0123456789;<=>?@ABCDEFGHIJKLMNPQRSTUUVWX 2..sgqt...sin cos tan ceil...floor..fabs...modf...ldexp...cabs...hypot.fmod...frexp...y0_y1_y2...yn_logb...n ticalSectionEx.CreateEventExW.CreateSemaphoreExW.SetThreadStackGuarantee.CreateThreadPoolTimer enCallbackReturns.GetCurrentProcessorNumber...GetLogicalProcessorInformation.CreateSymbolicLinkW.S rentPackageId.GetTickCount64.GetFileInformationByHandleExW.SetFileInformationByHandleW.U.S.E.R.3.2					

Viewing the raw log message generated by a Suricata IDS alert in the Security Onion SIEM.

Packet Captures

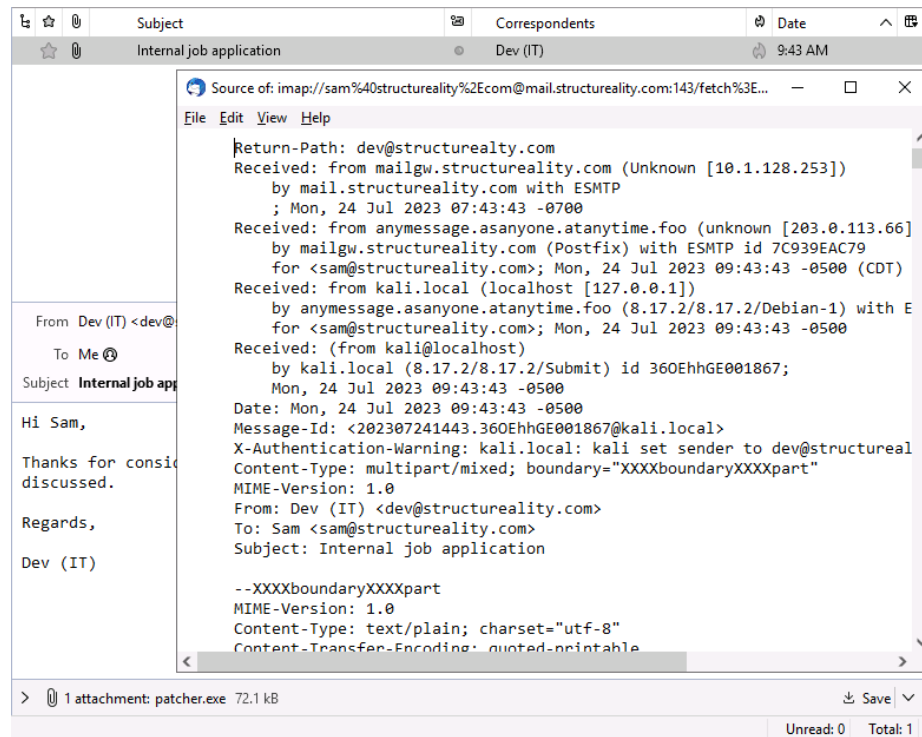


- Pivot from alert event to per-packet or frame analysis
- Extract binary data

Using the Wireshark packet analyzer to identify malicious executables being transferred over the Windows file-sharing protocol. (Screenshot Wireshark [wireshark.org](https://www.wireshark.org).)

Metadata

- File
 - Date/time and security attributes
 - Extended attributes and properties
- Web
- Email
 - Request and response headers
- Email
 - Internet header listing message transfer agents
 - Spam/security analysis



Analyzing headers in a phishing message: the sender is using typosquatting to hope the recipient confuses structureality.com with the genuine domain structureality.com.
(Screenshot courtesy of Mozilla.)

Review Activity: Data Sources

- Data sources, dashboards, and reports
- Log data
- Host operating system logs
- Application and endpoint logs
- Network data sources
 - Network appliance logs, firewall logs, IPS/IDS logs
- Packet captures
- Metadata

Lab Activity

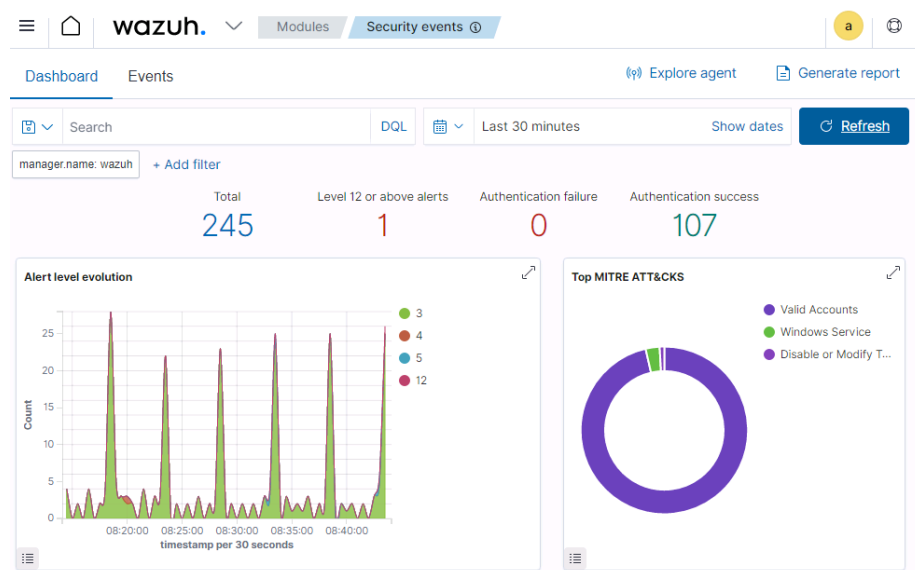
- Applied Lab: Using Network Sniffers

Lesson 12

Topic 12D

Alerting and Monitoring Tools

Security Information and Event Management



Wazuh SIEM dashboard—Configurable dashboards provide the high-level status view of network security metrics. (Screenshot used with permission from Wazuh Inc.)

- Log collection
 - Agent-based
 - Local agent to forward logs
 - Listener/collector
 - Protocol-based remote log forwarding (syslog)
 - Sensor
 - Packet capture and traffic flow data
- Log aggregation
 - Consolidation of multiple log formats to facilitate search/query and correlation
 - Normalization of fields
 - Time synchronization

Alerting and Monitoring Activities

- Alerting and correlation
 - Static rules and logical expressions
 - Threat intelligence feeds
 - Validation of alerts as true positives versus false positives
 - Quarantine for remediation
- Reporting
 - Executive, managerial, and compliance audiences
- Archiving
 - Preserve evidence of attack
 - Facilitate threat hunting and retrospective incident identification

Alert Tuning (1)

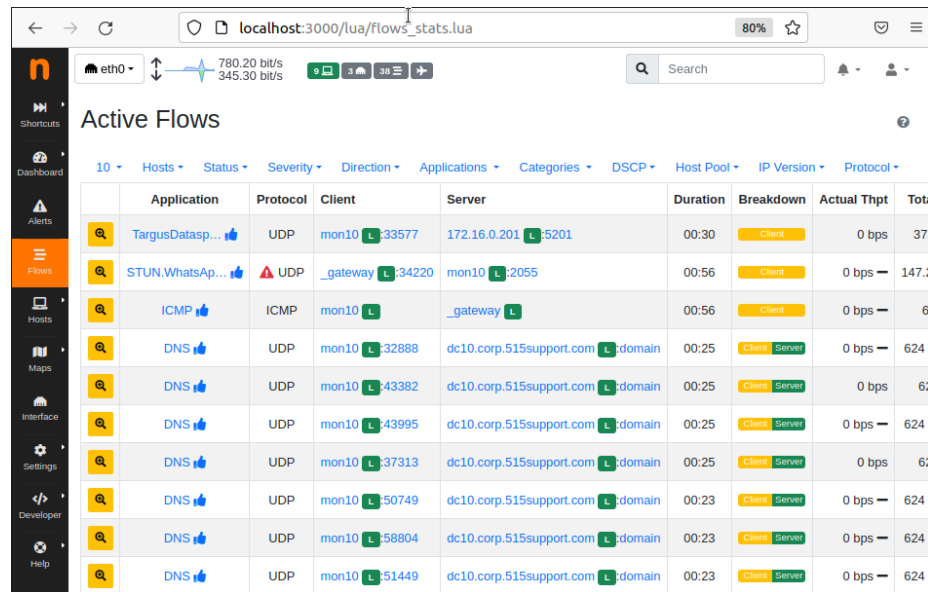
- Correlation action to log only, trigger alert, or trigger alarm
- Reduce false positives without increasing false negatives
 - False positives raise an alert when there is no actual malicious activity
 - False negatives do not raise an alert when there is malicious activity
 - True positives measure successful detection of incidents

Alert Tuning (2)

- Techniques for tuning
 - Refining detection rules and muting alert levels
 - Redirecting sudden alert “floods”
 - Redirecting infrastructure-related alerts
 - Continuous monitoring of alert volume and analyst feedback
 - Deploying machine learning (ML) analysis

Monitoring Infrastructure

- Network monitors
 - Appliance state data
 - Heartbeat availability monitoring
 - Simple Network Management Protocol (SNMP) traps
- Netflow/IPFIX
 - Records traffic statistics
 - Flows defined by endpoints and ports (keys)
 - Netflow exporters and collectors



	Application	Protocol	Client	Server	Duration	Breakdown	Actual Thpt	Tot
	TargusDataSp...	UDP	mon10 33577	172.16.0.201 5201	00:30	Client	0 bps	37
	STUN.WhatsAp...	UDP	_gateway 34220	mon10 2055	00:56	Client	0 bps	147.7
	ICMP	ICMP	mon10	_gateway	00:56	Client	0 bps	6
	DNS	UDP	mon10 32888	dc10.corp.515support.com domain	00:25	Client Server	0 bps	624
	DNS	UDP	mon10 43382	dc10.corp.515support.com domain	00:25	Client Server	0 bps	6;
	DNS	UDP	mon10 43995	dc10.corp.515support.com domain	00:25	Client Server	0 bps	624
	DNS	UDP	mon10 37313	dc10.corp.515support.com domain	00:25	Client Server	0 bps	6;
	DNS	UDP	mon10 50749	dc10.corp.515support.com domain	00:23	Client Server	0 bps	624
	DNS	UDP	mon10 58804	dc10.corp.515support.com domain	00:23	Client Server	0 bps	624
	DNS	UDP	mon10 51449	dc10.corp.515support.com domain	00:23	Client Server	0 bps	624

ntopng community edition being used to monitor NetFlow traffic data. (Screenshot used courtesy of ntop.)

Monitoring Systems and Applications

- System monitors and logs
 - System health reporting
 - System logs to diagnose availability issues
 - Security logs to audit access
- Application and cloud monitors
 - Application health monitoring
 - Cloud service health
- Vulnerability scanners
- Antivirus
- Data loss prevention (DLP)

Benchmarks

The screenshot shows the Wazuh dashboard interface. At the top, the 'wazuh.' logo is visible, followed by a dropdown menu and a 'Modules' tab. The 'NIST 800-53' module is selected. Below the navigation bar, there are tabs for 'Controls', 'Dashboard', and 'Events'. The 'Controls' tab is active. A search bar is present, and a filter is applied: 'manager.name: wazuh rule.nist_800_53: exists'. The main content area displays a table of requirements. On the left, a sidebar shows the 'NIST 800-53' framework with a list of requirements and their counts: 'Requireme...' (163), 'Requireme...' (9), 'Requireme...' (0), 'Requireme...' (0), and 'Requireme...' (0). The main table is titled 'Requirements' and has a 'Filter requirements' search bar. It contains a grid of requirement cards, each with a title and a count. The requirements are organized into four columns and five rows. The first row contains: 'AC.7 - UNSUCCESS...' (163), 'AU.6 - AUDIT REVIE...' (7), 'AU.5 - RESPONSE T...' (2), and 'AC.2 - ACCOUNT M...' (0). The second row contains: 'AC.6 - LEAST PRIVIL...' (0), 'AC.12 - SESSION TE...' (0), 'AU.8 - TIME STAMPS...' (0), and 'AU.9 - PROTECTION ...' (0). The third row contains: 'AU.12 - AUDIT GENE...' (0), 'CA.3 - SYSTEM INTE...' (0), 'CM.1 - CONFIGURATI...' (0), and 'CM.3 - CONFIGURAT...' (0). The fourth row contains: 'CM.5 - ACCESS RES...' (0), 'IA.4 - IDENTIFIER M...' (0), 'IA.5 - AUTHENTICAT...' (0), and 'IA.10 - ADAPTIVE IDE...' (0). The fifth row contains: 'SA.11 - DEVELOPER ...' (0), 'SC.2 - APPLICATION...' (0), 'SC.7 - BOUNDARY P...' (0), and 'SC.8 - TRANSMISSI...' (0). A toggle switch 'Hide requirements with no alerts' is located at the top right of the table.

Requirement	Count	Requirement	Count	Requirement	Count	Requirement	Count
AC.7 - UNSUCCESS...	163	AU.6 - AUDIT REVIE...	7	AU.5 - RESPONSE T...	2	AC.2 - ACCOUNT M...	0
AC.6 - LEAST PRIVIL...	0	AC.12 - SESSION TE...	0	AU.8 - TIME STAMPS...	0	AU.9 - PROTECTION ...	0
AU.12 - AUDIT GENE...	0	CA.3 - SYSTEM INTE...	0	CM.1 - CONFIGURATI...	0	CM.3 - CONFIGURAT...	0
CM.5 - ACCESS RES...	0	IA.4 - IDENTIFIER M...	0	IA.5 - AUTHENTICAT...	0	IA.10 - ADAPTIVE IDE...	0
SA.11 - DEVELOPER ...	0	SC.2 - APPLICATION...	0	SC.7 - BOUNDARY P...	0	SC.8 - TRANSMISSI...	0

Monitoring template aligned to NIST 800-53 framework requirements.

- Scanning for configuration vulnerabilities
 - Lack of controls
 - Improper configuration
- Security content automation protocol (SCAP)
 - Language to enable scanners to load configuration benchmarks and scan for deviations

Review Activity: Alerting and Monitoring Tools

- Security Information and Event Management
 - Collection and aggregation
- Alerting and monitoring activities
 - Correlation, reporting, archiving
- Alert tuning
 - False positives, false negatives, alert fatigue
- Monitoring infrastructure
 - Network monitors, NetFlow
- Monitoring systems and applications
- Benchmarks
 - SCAP, OVAL, XCCDF configuration baseline scanning

Lab Activity

- Assisted Lab: Performing Root Cause Analysis

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Lesson 12



Summary